Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2008
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2010 (from 5/31/2009) per PI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Herlach, Dieter (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation)		
Grant/Contract No.:	NNX07AK27G		
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Performance Goal Text:			

Task Description:	In 1724, Fahrenheit first observed that liquid water could be cooled by a significant amount below its freezing temperature before crystalline ice forms. In 1952, David Turnbull showed that this ability to supercool was a common property of liquids, occurring even in metallic liquids. His experiments demonstrated that a barrier exists to the initial formation of the crystal from the liquid, a process called nucleation. This process is the starting point of not only crystallization, but of most phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. The attainment of an improved ability to control nucleation is of central importance for the development of new materials for advanced applications. In almost all cases the classical theory of nucleation, first developed in the late 19th century, is used to analyze and predict nucleation results. There are, however, serious problems with aspects of that theory in significant regimes of its application. Further, it does not properly describe nucleation when the concentrations of the initial and final phases are different. The task of the continuing ground-based research on this contract is the investigation of the icosahedral ordering and associated thermophysical properties in supercooled liquids, primarily of transition metals and their alloys. Measurements are being made of the structure, specific heat, density, viscosity, surface tension and emissivity in the liquids. These data, when combined with the nucleation data, will lead to a better understanding of the complicated nucleation processes that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
Task Progress:	In 2005 NASA decided to put on hold planned ISS experiments, including our flight experiment, QUASI. We were allowed a phase-out period for that contract, NNM04AA01G. However, recognizing the importance and visibility of our ground-based research, and the potential for a successful ISS experiment should a future opportunity arise, we were issued a new contract, NNX07AK27G to continue our ground-based studies. The analysis of existing thermophysical and structural data for supercooling liquids obtained at the Advanced Photon Source was supported by residual funds in NNM04AA01G. The results from the past year are summarized below. Some of these studies were also relevant to related work under our NSF grants, DMR 0307410 and DMR 0606065, studying the relations between structure and ordering in the liquid and the glass and the influence of this ordering on glass formation, the glass transition and the crystal nucleation barrier. 1. Previously, we demonstrated the development of icosahedral short-range order (ISRO) in equilibrium and supercooled Ti-Zr-Ni liquids. In-situ high-energy x-ray diffraction data were obtained from electrostatically-levitated liquids using the Beamline Electrostatic Levitation (BESL) method that was developed under this contract. Coupled with nucleation data on the same liquid, these studies were the first demonstration of a half-century hypothesis relating the liquid structure to the nucleation barrier. Since the last report, we have extended those studies to show that from a cluster-based analysis of the X-ray structure factors the ISRO in the alloys becomes progressively more dominant, and the coherence length of the order becomes longer, with the addition of Ni, especially near a concentration of 21 at.% Ni. 2. Undercooling experiments on a large number of simple crystalline, polytetrahedral, and icosahedral quasicrystalline phase forming compositions in Ti-Zr-Ni alloys were carried out using electrostatic levitation (ESL) techniques for containerless processing. Consistent with Frank's hypothesis, a direct correlation was found between the reduced undercooling and the icosahedral order in the solid. The reduced undercooling is less for liquids that form the icosahedral quasicrystal (i-phase) than for those that form the hcp C14 Laves polytetrahedral phase. For many compositions near 21 at.% Ni, the primary nucleation of a metastable i-phase instead of a stable C14 Laves phase demonstrates that the interfacial free energy between the liquid and the i-phase is smaller than between the liquid and the C14 Laves phase, indicating icosahedral local order in the undercooled liquid. This is in agreement with a classical-nucleation-theory-based estimate of the interfacial free energy and the work of formation of the critical cluster from the undercooling data. 3. Our earlier conclusions about ISRO were based on the assumption of a single dominant cluster type in the liquid and the observed evolution of a high-q shoulder on the second peak in the structure factor, S(q). New diffraction data for Ti37.5Zr37.5Ni21 were obtained under the support of this contract, using more rapid data acquisition methods. These allowed structural studies to be made down to and through recalescence to the icosahedral quasicrystal. The liquid atomic structures obtained from a Reverse Monte Carlo analysis of these data were characterized by their bond angle distributions, Honeycutt and Andersen indices and bond orientational order parameters. The studies showed that while a single cluster was not able to describe the data, the dominant order was icosahedral, supporting our earlier conclusions. 4. Earlier studies by us, assuming a dominant cluster in the liquid, indicated that the ISRO in early transition liquids (Ti and Zr) was distorted, compared to that in late transition metal liquids such as Ni. To check this further, the atomic structures of the liquids were determined from a RMC analysis of the diffraction data. These were analyzed in terms of their bond angle distributions, Honeycutt-Anderson indices and bond orientational order parameters. Contrary to our previous single-cluster studies these results indicate that there is clearly no single cluster type that fully describes these liquids. However, in limited agreement with the previous cluster-based studies of the diffraction data, supercooled liquid Ti develops a significant (though not exclusive) amount of distorted icosahedral short-range order with supercooling, while the short-range order in supercooled liquid Zr and Ni is less distorted. The short-range order in Ni is particularly close to that of an undistorted icosahedron. This difference in the evolution of the local structures of the early and late transition metals likely reflects a competition between close packing and the angular dependence of the atomic potential. 5. The local structure of supercooled liquid Zr67Pd33 was studied using BESL The distinct shoulder on the high-q side of the second peak in the structure factor indicated the development of ISRO with supercooling. The atomic structures obtained by a Reverse Monte Carlo simulation of the experimental data were analyzed in terms of their Honeycutt and Andersen indices and bond orientational order parameters (BOO) to confirm this. These studies indicate that Zr-centered icosahedral clusters exist in the supercooled liquid and that the fraction of the icosahedral clusters increases with supercooling.

	6. Liquid silicon is known to be metallic while the diamond cubic solid form is a well-known semiconductor. Theoretical treatments have argued for a liquid/liquid phase transition in the supercooled state at which the local order changes from the metallic A5 type to the semiconductor A4 one. To investigate this prediction, we have constructed the atomic structures of supercooled liquid silicon from BESL X-ray diffraction data obtained from the levitated liquid. A Reverse Monte Carlo method was used to determine the atomic structures as a function of supercooling. Based on a Honeycutt-Anderson and bond orientational analyses of these structures, the supercooled liquid silicon maintains the A5 structure (with some A4), with the coherence length of the A5 order increasing with supercooling. The ordering of the A5 structure is consistent with previous experimental observations, such as a constant coordination number, constant nearest-neighbor distances and the evolution of the first peak in S(q) and of the second peak in g(r) with supercooling. These observations are in opposition to the theoretical predictions of a liquid-liquid structural phase transition toward a diamond cubic structure (characteristic of crystal Si) in supercooled liquid Si, Further, they indicate that baring the intervention of phase transitions such as crystallization, liquids will order towards a preferred structure that is characteristic of the liquid state and the nature of the atomic bonding.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Kim TH, Lee GW, Gangopadhyay AK, Hyers RW, Rogers JR, Goldman AI, Kelton KF. "Structural studies of a Ti-Zr-Ni quasicrystal-forming liquid." Journal of Physics: Condensed Matter 2007 Nov 14;19(45):455212(10 pp). http://dx.doi.org/10.1088/0953-8984/19/45/455212 , Nov-2007
Articles in Peer-reviewed Journals	Kim TH, Goldman AI, Kelton KF. "Structural study of supercooled liquid silicon." Philosophical Magazine 2008;88(2):171-9. http://dx.doi.org/10.1080/14786430701798944 , Jan-2008